Education and Information Technologies (2025) 30:18749–18778
https://doi.org/10.1007/s10639-025-13516-6

From text to moving image: Evaluating generative artificial intelligence text-to-video models for pre-writing idea generation in language instruction

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Received: 25 June 2024 / Accepted: 19 March 2025 / Published online: 3 April 2025

Abstract

Research on using generative artificial intelligence (GenAI) chatbots in learning environments continues to underpin research in and out of academia. However, GenAI text-to-video (T2V) models have not been adequately investigated in academia, mostly due to their novelty. That means their potential to improve learning remains to be determined. This study contributes to bridging this gap by investigating how GenAI T2V models can foster idea generation during brainstorming. The participants are two groups (experiment and control) of college freshmen, with one group presented with AI-generated videos during brainstorming. Data gathered from the participants' screen recordings, individual notes, a questionnaire, and an assessment of brainstormed ideas to determine whether the participants in the experiment group benefited from the background and narrative elements of the AI-generated videos showed that more ideas were generated within a shorter time, and the videos were perceived to be useful and of high quality. Though the findings did not overwhelmingly support an empirically informed case for deploying GenAI T2V models in descriptive writing, they have significant implications for teachers, students, and institutions of higher learning.

Keywords Generative artificial intelligence (GenAI) · Text-to-video (T2V models) · Descriptive writing · Brainstorming · Idea generation

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1 Introduction

The advent of generative artificial intelligence (GenAI) has disrupted teaching and learning and opened up numerous possibilities to improve learning outcomes. One area wherein technology has made a significant impact is reading and writing, particularly the realm of idea generation. Researchers (Isaksen & Gaulin, 2005) have identified brainstorming (BS) as a particularly promising area where technology can overcome inherent ideation challenges. However, the focus of research in academic writing has largely been on the impact of GenAI chatbots, with ChatGPT being the most investigated artificial intelligence (AI)-based tool (AlGhamdi, 2024; Filippi, 2023; Meron & Araci, 2023). Despite this growing interest, much remains to be explored regarding the specific applications of GenAI beyond chatbots. The potential of GenAI chatbots, especially in academic writing, is still a subject of interest and investigation. As these tools become more prevalent, both learners and instructors will continue to explore the possibilities they offer.

Nevertheless, a perusal of the extant literature shows that BS, an essential step in the writing process, has not significantly benefited from investigations into AI's usefulness in writing. For instance, Niloy et al. (2024) assessed ChatGPT's role in the writing process but overlooked BS to focus on content originality, presentation, accuracy, and elaboration. Meanwhile, Chukwuere (2024) adopted a holistic approach focusing on AI's role as "writing support, data analysis, literature review, and scientific cooperation" (p. 30). Empirical evidence is also scant for AI's application to writing while focusing on helping learners brainstorm ideas when writing different types of essays, including descriptive essays. Yu-Han and Chun-Ching (2023) found that students who brainstormed together with AI produced more substantial ideas than their peers who did not receive AI support. Additionally, Karanjakwut and Charunsri (2025) investigated how AI-driven BS tools impacted the writing process and students' perceptions. Though the experimental class outperformed the control class, concerns about overdependence on AI persisted. Furthermore, instructors tended to prefer ideas generated by the control class, perceiving them as more authentic. Several recent studies on the role of ChatGPT in enhancing writing have either been prescriptive (AlAfnan et al., 2023), grounded in interviews and questionnaires (Sudrajad et al., 2024) rather than data generated during actual writing, or literature review syntheses (Imran & Almusharraf, 2023). Recent advances in AI development, especially GenAI text-to-video (T2V) models, deserve more attention within the academic writing context, given that the models possess unique idea-generation potential during BS that warrants exploration. However, investigations on how they can be employed by learners in classrooms are scarce. Even more elusive are students who adopt a process-oriented methodological framework that highlights concrete usage and practices. Given the models' idea generation potential during BS, experiments on learners' practices, perceptions, and brainstormed ideas are crucial in informing pedagogical strategies. Thus, this study aims to bridge these gaps by examining whether and how AI-generated videos can foster idea creation during BS.

The study participants were freshmen enrolled in an academic writing course at a Chinese university. Two classes were investigated, one serving as the experimental group (EG) and the other as the control group (CG). Each group comprised 23 participants. The data obtained from a questionnaire, the participants' mind maps, personal notes, brainstormed ideas, and screen recordings of the BS process were analyzed to determine whether the EG generated more ideas during BS, benefited from the GenAI T2V's background and narrative elements, spent less time BS, and had a positive perception of the video's quality and usefulness. By addressing these research questions, this study sought to provide valuable insights into the applicability of AI in academic writing, particularly in the context of BS, and to contribute to the ongoing discourse on integrating technology in foreign language (FL) instruction.

The study is significant in several dimensions: First, it fills critical gaps in the literature regarding the role of BS in academic writing using GenAI tools. Second, the focus on AI-generated T2V models in facilitating idea generation during BS unveils a pedagogical strategy for teaching writing that improves the learners' abilities to generate original ideas. As GenAI technologies become increasingly prevalent in educational settings, understanding their applications in specific contexts is crucial for guiding effective technology integration. Third, the study has the potential to inform educators and policymakers about the practicalities of incorporating AI into FL instruction, paving the way for future research on innovative approaches to teaching and learning. Overall, this research fosters important discourse surrounding the applicability of AI in education and contributes valuable insights that can influence both practice and further exploration in the field.

2 Literature review

2.1 Artificial intelligence and writing

The volume of literature on AI in writing at various levels continues to increase exponentially with the democratization of GenAI. As a natural consequence, researchers have focused on AI's role in improving various types of writing, including scientific, technical, formal, and informal writing. From descriptive to empirically informed research, the consensus is that GenAI has the potential to improve scientific writing. Salvagno et al. (2023) speculated that GenAI may "help in the literature review, identify research questions, provide an overview of the current state of the field, and assist with tasks, such as formatting and language review" (p. 4). In more empirically founded research, Safrai and Orwig (2024) utilized ChatGPT-4 to generate a review of fertility preservation in men and prepubertal boys. They concluded that ChatGPT-4 successfully created a review that included 36% accurate, 48% partially correct, and 16% purely fabricated references. However, though the experts rated the article as highly relevant, it had a low score in depth and currentness. In a study on using GenAI in writing biomedical reports, Kumar (2023) posited that despite its multiple limitations, GenAI "has enormous potential as training and upskilling resource for academic writing, which rather than replacing biological intelligence will help in refining it if used appropriately under academic mentoring" (p. 30), thus capturing the essence of current investigations of GenAI in scientific writing.

Similar conclusions have been reached regarding the employment of GenAI chatbots in technical writing. For example, AlAfnan et al. (2023) utilized ChatGPT to generate theory-based ideas for business communication. Though the chatbot extracted information verbatim from online sources, the authors found that "the answers/responses to the theory-based question were correct, and they received high grades" (AlAfnan et al., 2023, p. 63). While empirically informed AI-based studies remain sparse in technical writing, optimism regarding the technology's potential continues. According to Card and Duin (2023), the ubiquity of GenAI means that technical writers need to consider humans and AI in their creation of technical documentation, particularly with regard to autonomous authoring.

Most studies on GenAI and writing have focused on academic and research paper writing. Researchers have evaluated students' perceptions of GenAI chatbots' usefulness in writing, and their conclusions have shed light on learner expectations vis-à-vis the tool. For instance, Dong (2024) found that learners' performance expectations influence their perception of and intention to use ChatGPT. Other studies (Zebua & Katemba, 2024) have primarily uncovered positive impressions among students who have benefited from using GenAI chatbots in writing tasks. Moreover, empirical evidence from multiple investigations has revealed that GenAI improves writing performance and skill development (Mahapatra, 2024), can save time (Ariyaratne et al., 2023), and can be particularly useful in polishing academic writing, improving peer reviews, and optimizing feedback (Gruda, 2024). Beyond investigations on general academic concepts, such as performance and skill development, researchers have also focused on specific elements of writing, foregrounding GenAI's assistance in the writing process (Punar Özçelik & Yangın Ekşi, 2024; Zou & Huang, 2023), prompt engineering (Giray, 2023), and workflow automation (Yan, 2023).

However, research has also revealed multiple challenges related to using GenAI in academic writing. Some studies (Mennella & Quadros-Mennella, 2024) have found negligible differences between students writing with and without the help of GenAI. For instance, after experimenting with two groups of students, including one group relying on ChatGPT-3.5, Bašić et al. (2023) concluded that "the ChatGPT group did not perform better in either of the indicators; the students did not deliver higher quality content, did not write faster, nor had a higher degree of authentic text" (p. 1). Furthermore, several researchers have raised concerns about plagiarism (Yan, 2023) and GenAI chatbots' inability to provide learner-tailored responses (AlGhamdi, 2024).

2.2 Generative artificial intelligence and brainstorming

BS is a crucial component of narrative and descriptive writing, where learners utilize storytelling and vivid descriptions to engage readers in multiple ways. Researchers have found that GenAI can assist learners during BS. In an experiment with ChatGPT, Lingard (2023) concluded that “ChatGPT can be a good brainstorming resource” (p. 266) while cautioning users to be mindful of hallucinations, bogus information, and ethical questions regarding its use. Meron and Araci (2023) found that ChatGPT is “a useful tool for brainstorming, as well as structuring and editing language, potentially saving time” (p. 21). Yu-Han and Chun-Ching (2023) discovered a positive and significant correlation between participants’ confidence levels and the chatbot’s suggestions, implying that AI can extend human thinking in BS. In a study by Lavrič and Škraba (2023), ChatGPT-3.5 supported a group of learners during BS, with the research focusing on the distinction between human-generated and artificially generated ideas and how such a combination could enhance ideation. Meanwhile, Filippi (2023) investigated ChatGPT’s role in fostering ideation in product design and asserted that though the AI chatbot is not particularly useful in suggesting concepts, it “performed quite well, contrary to expectations since its knowledge base contains only pieces of information regarding things that have already happened (and been documented) in the past” (p. 14). Other studies foregrounding the usefulness of GenAI in BS have been undertaken in content marketing (Schmidt et al., 2023) and, less conspicuously, in academic writing.

2.3 Existing lacunae

Despite the extensive research conducted on AI and writing, a review of the extant literature unveiled significant gaps. First, insufficient research has been undertaken on the usefulness of AI in BS, a foundational activity in writing. Precisely, little empirical data informs GenAI’s usefulness (or lack thereof) in idea generation within the descriptive and narrative writing context. Second, though research indicates that ChatGPT can suggest ideas during BS (Tsufim & Pomerleau, 2024), scholars have not adequately explored the chatbot’s ability to generate narrative and descriptive writing ideas within a college English as a foreign language (EFL) scenario, especially using a process-oriented methodology. This means that investigations into how students obtain and modify ideas and their individual interactions with the chatbot remain elusive.

Furthermore, GenAI chatbots appear to be the most investigated AI tool in writing. Presumably, due to its ubiquity and popularity, ChatGPT is the most explored GenAI in college writing. Other AI-based chatbots, including Google’s Bard, Microsoft’s Copilot, and Baidu’s Ernie Bot, have hardly been the focus of much investigation. Most importantly, recent GenAI T2V models, such as OpenAI’s Sora and Fliki (Zhou et al., 2024), have rarely been explored within descriptive writing idea generation.

GenAI T2V models: This technology allows users to provide a text, ranging from a simple summary to a detailed series of scenes. Powerful AI models are employed to analyze the text and identify key information, including characters, actions, settings, and overall tone. Based on the information extracted, the AI generates corresponding visuals, such as images and animations, accompanied by transitions, narration, and background music. GenAI T2V models are novel, and preliminary research (Sun et al., 2024; Zhou et al., 2024) indicates that GenAI T2V, though inaugural, "has dramatically reignited optimism, marking a momentous shift and breathing new life into the momentum of the field" (Sun et al., 2024, p. 4). Meanwhile, in a study on users' perceptions of Sora, OpenAI's T2V software, Liu et al. (2024) concluded that the model is significantly disruptive in multiple industries, including the cinema, gaming, healthcare, and robotics fields. Regarding education, Zhou et al. (2024) argued that Sora may "become a potentially powerful tool..., offering opportunities for diverse learning modalities, creativity, and critical thinking" (p. 1).

Therefore, this study contributes to the discourse on GenAI T2V by focusing on the technology's usefulness in education. The research focused on the role of an AI-based tool in assisting EFL learners in generating ideas during BS. Conducted in a learning environment, the study answered the following research questions (RQs):

RQ1: Did the GenAI T2V models foster idea creation during BS?
RQ2: Which elements of the AI-generated videos were most explored by the learners?
RQ3: What were the learners' perceptions regarding GenAI T2V?

The answers to these questions provided relevant material that informed our reflections on the practical and pedagogical implications of GenAI T2V models in academic writing.

3 Methodology

This study was conducted using a mixed methods approach, which yielded data obtained from a questionnaire and the participants' notes, screen recordings, and ideas generated during BS. This section presents the participants, study design, data collection and instruments, and data analysis.

3.1 Participants

The participants were freshmen majoring in English at a Chinese university. Initially, 53 students from two classes (control and experimental) participated in the study. However, eight participants were eliminated because they failed to record their screens adequately or did not participate in the first (BS1) or the second (BS2) tasks. Only the data from 43 participants (CG: n = 23; EG: n = 23) was analyzed. The EG included 19 female and 4 male participants, while the CG included 17 female and 6 male participants. They were admitted to study English based on their college entrance examination scores and an internal university assessment, based on which the two classes were ranked at an identical proficiency level. Therefore, we saw no need to assess their proficiency level further before conducting the study. The participants brainstormed using an e-BS tool. However, while the CG brainstormed traditionally (using a mind map), the EG was provided with an AI-generated video about the topic and was then requested to brainstorm ideas. This study assessed how the AI-generated video contributed to idea creation during the BS.

The participant sample size, though seemingly small, was representative and consistent with previous studies on writing in college. Multiple studies have examined GenAI chatbots' usefulness without assessing students' writing (AlAfnan et al., 2023; Ariyaratne et al., 2023; Giray, 2023) or have focused on personal experience (Gruda, 2024) with the tool. Other researchers have assessed students' writing using qualitative methods (AlGhambi, 2024) or questionnaire data (Dong, 2024). By contrast, studies based on in-class experiments focusing on AI-supported writing have included few participants. For example, Escalante et al. (2023) assessed 48 participants in two groups, and Bašić et al. (2023) had 18 participants (9 in each group) in their study.

3.2 Study design

The study was conducted in week 5 of an 18-week-long semester, during which the participants learned how to write descriptive essays based on the plan in Appendix 1. A pretest was administered in Week 6 to evaluate baseline knowledge of BS. No statistical difference was found between the two groups. The BS task was in the seventh week after the participants had been introduced to descriptive essays and learned how to brainstorm individually and in groups. They practiced "thinking more and writing less"—that is, using single words, phrases, and clauses to represent ideas developed during the writing stage. In addition, they learned how to download, install, and use EV Recorder, a popular screen recording software utilized in related investigations (Tekwa, 2024). This way, they familiarized themselves with the recording tool and the BS practice before the experiment. Problems related to the recording tool, BS process, and the participants' devices (personal computers and tablets) used during the study were resolved during the preparation weeks. To ensure consistency, both groups participated in the study under the same environmental conditions, including time of day and location. The CG had their first two lessons in the morning, followed by the EG in the same classroom and under the same conditions.

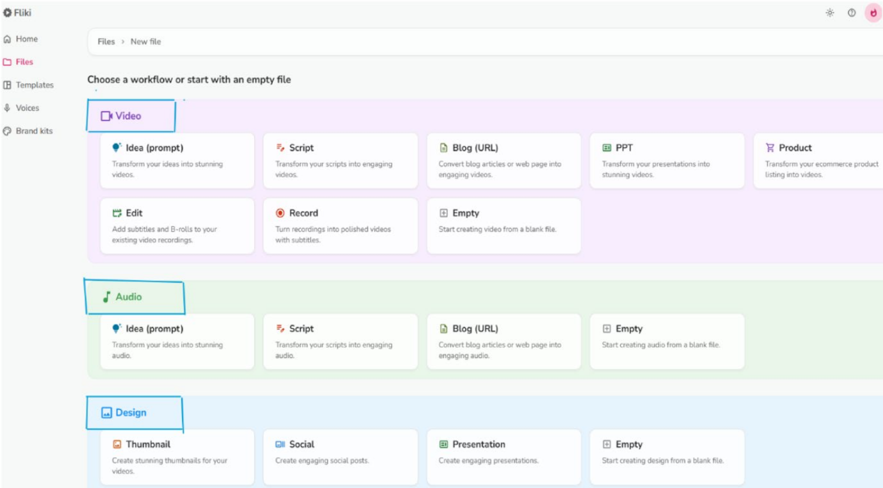
3.3 Data collection and instruments

The study's data were gathered from the participants' notes, mind maps, brainstormed ideas, screen recordings, and questionnaire answers. This section presents how the data were collected using various instruments and describes the BS process.

3.3.1 Artificial intelligence–generated videos

Two essay topics, "My 20th birthday" and "A beach party I organized," were independently introduced to the AI T2V software, which then created two videos. Fliki, the AI-based T2V software used here, allows users to generate a landscape or portrait video, audio, and a design file from an idea, a script, a blog, PowerPoint slides, or a tweet, as presented in Fig. 1 (Screenshot 1). Users can also choose their

Screenshot 1



Screenshot 2

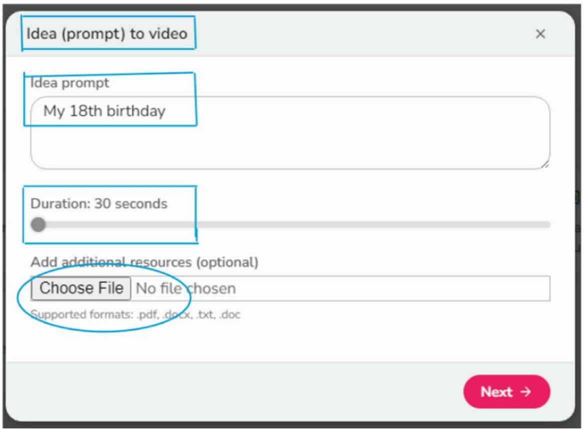


Fig. 1 Screenshot of Fliki, the artificial intelligence–based text-to-video software

language (dialect), specify the video length, provide a prompt, and select videos or images to generate videos, as outlined in Screenshot 2.

The AI-based software generates text and selects videos to accompany each idea. Users can modify the text and videos by searching and selecting from an existing stock on the platform's library. Furthermore, subscribed users can create personalized libraries, with videos uploaded from their laptops and stored in Fliki's online library.

Therefore, two videos corresponding to the aforementioned essay topics were generated for the experiment. First, each topic was independently introduced into the AI T2V software, which generated a video (images and voice narration). The

Video 1 background elements

Element 1:	Multiracial people
Element 2:	Hand in hand
Element 3:	Burning candles
Element 4:	Red/blue ribbons
Element 5:	On-going party
Element 6:	Drinks
Element 7:	People smiling
Element 8:	Close friends and families
Element 9:	Richly decorated cake
Element 10:	Beads
Element 11:	Birthday wishes
Element 12:	People talking excitedly
Element 13:	Evening
Element 14:	Twinkling lights
Element 15:	Picture album
Element 16:	Experiencing new things in life
Element 17:	People at a beach

Video 2 background elements

Element 1:	Sunset
Element 2:	Sandy beach
Element 3:	People dancing
Element 4:	People in swimsuits
Element 5:	T-shirt and jeans
Element 6:	Young people together
Element 7:	Sea waves
Element 8:	People laughing and having fun
Element 9:	Invitation cards
Element 10:	Stars and the sun
Element 11:	Boys and girls playing in water
Element 12:	Tropical flowers
Element 13:	DJ playing music
Element 14:	People partying all night
Element 15:	Adult men drinking beer
Element 16:	BBQ
Element 17:	People laughing out loud
Element 18:	Fire and fire dancers
Element 19:	Moon light

Fig. 2 Background elements extracted from the artificial intelligence–generated Video 1 and Video 2

Video 1 narrative elements

Element 1:	A day to remember
Element 2:	Sun in horizon
Element 3:	Perfect scene
Element 4:	Unforgettable birthday
Element 5:	Family and friends
Element 6:	Buzzes with excitement
Element 7:	Heartfelt laughter
Element 8:	Evening unfolds
Element 9:	Twinkling lights and music
Element 10:	Magical moments
Element 11:	Shared stories
Element 12:	Creating memories to cherish
Element 13:	Highlights of the event
Element 14:	Splendid cake
Element 15:	Mirrors the life I have journeyed through
Element 16:	Not only for myself, but for ...
Element 17:	Cheers to a new chapter
Element 18:	Love and endless possibilities
Element 19:	Milestone

Video 2 narrative elements

Element 1:	Sunshine kisses your skin
Element 2:	Sea waves serenade you
Element 3:	Idyllic spot
Element 4:	Golden sands
Element 5:	Crystal clear waters
Element 6:	Unforgettable day
Element 7:	An escape into paradise
Element 8:	Guests
Element 9:	Transformation
Element 10:	Tables adorned with tropical flowers
Element 11:	DJ booth
Element 12:	Aroma of BBQ and sea salt
Element 13:	Illuminate the night
Element 14:	Outbursts of laughter
Element 15:	Zenith
Element 16:	Invitation

Fig. 3 Narrative elements extracted from the artificial intelligence–generated Video 1 and Video 2

software was then instructed to create subtitles for the videos by converting the voice into text. Then, the participants watched the videos, listened to the text, and read the subtitles. Each one-minute-long video contained background and narrative elements (Fig. 2) to trigger ideas during BS. Both AI-generated videos were designed to be equivalent in quality and complexity to control for any potential bias.

In addition to the background elements, the narrative elements outlined in Fig. 3 were extracted from the audio and subtitles of each video.

Fig. 4 Layout of the Mubu e-mind mapping tool

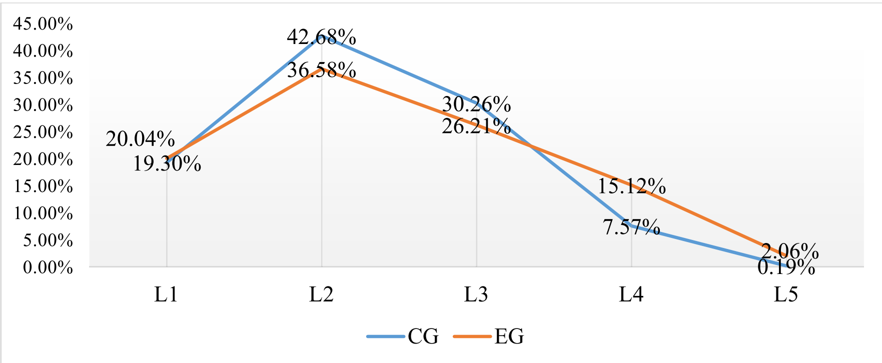


Fig. 5 Percentage of ideas generated by the control group (CG) and experimental group (EG) per brainstorming level

The narrative elements included single words (adjectives and nouns), noun phrases, clauses, and other expressions considered of semantic and lexical importance to understanding the video. The videos' narrative and background elements were selected in consultation with two Western Movie course instructors with a combined 33 years of film studies experience (Fig. 4).

3.3.2 Brainstorming process and generated ideas

Mind mapping was the primary BS technique employed in the study, and the participants used Mubu-4, an e-mind mapping tool (Fig. 5) designed to be similar to MindMeister and Coggle. A nominal BS format was employed using an e-BS tool, making it possible to generate infinite ideas. Three main BS rules were provided in the study, as follows: quantity (i.e., generation of as many ideas as possible, with[out] connections or practical relations among them); freewheeling (i.e., no need for a connection or relationship among the various ideas generated); and non-evaluation (i.e., no checking or evaluation of the ideas generated) (Osborn, 1957). The participants were given approximately 15 min for the task, and they were required to generate as many ideas as possible while recording their screens. The e-BS tool was a mind map with the topic at the center; Level 1 ideas branched off from the topic to Level 2, Level 3, Level 4, and Level 5 +, as presented in Fig. 4.

The more levels generated, the more details the participants were presumed to provide about the topic. The instructions provided to the participants were standardized to ensure consistency in task execution across both groups. During the activity, a trained MA student assistant helped to verify that participants followed the same procedure to maintain consistency. No inconsistencies were reported, given that the participants were already very familiar with the BS process.

3.3.3 Participants' notes

The participants made notes shortly after completing the BS task. They were required to watch their screen recording to ensure the entire activity was recorded and then identify moments of inactivity (i.e., when their cursors did not move) and write down what they were doing. This technique, known as recall memory (Cooke et al., 1986), measures explicit memory and focuses on bringing back or remembering previously known information. This allowed for the moments during the BS when the participants appeared to do nothing to be accounted for. First, the participants wrote notes in either their notebooks or on their tablets, smartphones, or personal computers. Then, they transferred the notes onto an online cloud-based Excel document for easy download and analysis. The participants were instructed to complete their notes immediately after the BS session to mitigate recall bias. In total, 1,202 words of notes were obtained and analyzed to explain the participants' unrecorded activities.

3.3.4 Questionnaire

The participants' perceptions of the AI-generated videos were obtained using a questionnaire adapted from tested and validated questionnaires on AI-generated feedback (Escalante et al., 2023), technology acceptance models' perceived usefulness (Davis, 1993), and subjective video quality (ITU, 2008; an assessment survey developed by the International Telecommunication Union). Comprising predominantly Likert scale–type questions, the questionnaire contained 15 questions that gathered information on the participants' perceptions of the video quality, generated content, and narrative elements. It measured the extent to which the participants found the content, background, and narrative elements of the AI-generated videos to be central to their BS endeavor.

The questionnaire was designed on the FreeOnlineSurvey platform (www.freeonlinesurvey.com), which facilitated the design and dissemination of the surveys and their conversion to numerals and collection. A pilot test of the questionnaire was conducted to identify and eliminate potential biases in the questions and to ensure question clarity. A link and QR code were generated and disseminated to the participants via the class WeChat group. WeChat is a popular social media and instant messaging platform that facilitates the easy sharing of different file types. Various studies have documented its contribution to research (Jin, 2018). All participants completed the questionnaire in class within an average of 64 s. Their responses were then downloaded in numerical format into Excel files and analyzed accordingly.

3.4 Data analysis

The data were analyzed to answer the RQs and sub-questions. To answer RQ1 (i.e., to determine whether AI-generated videos fostered ideation during BS), VNote (Tekwa, 2024) video analysis software was utilized to code the screen recordings. Various color codes were used to determine the total time spent on the activity and the real BS time for individuals and groups. Then, individual participants' ideas were tallied to obtain the total ideas (mean [M] score and standard deviation [SD]) generated by the CG and EG. In this study, the focus was to determine how many ideas the participants generated. Therefore, in creating thematic and sub-thematic categories based on the levels of the BS ideas, care was taken by the two researchers working together to ensure that no redundant or repeated ideas were included in the tally of each student. In terms of diversity and usefulness, we evaluated how well they aligned with the GenAI T2V's background and narrative features.

The data of the two groups were compared using an independent sample t-test, which either confirmed the statistical differences between the two datasets (p-value of 0.05 or less) or did not confirm the statistical differences between them. Where differences existed, the analysis unveiled practical differences determined by Cohen's d value.

Further comparisons between the ideas generated by the two groups were undertaken by analyzing the levels with the most generated ideas. The information was obtained by calculating the M and SD per level of ideas generated. The finding showed discrepancies between the two groups, obtained by investigating the impact of the AI-generated videos. Meanwhile, to obtain the actual BS time, the mean scores of the actual BS time were subtracted from the total BS time. The correlation between actual BS time and total ideas generated was calculated to determine the relationship between the two variables for both classes.

To answer RQ2, we compared whether the ideas generated by the participants aligned with the AI-generated videos' background and narrative elements. To this end, each video's background and narrative elements were manually listed. Then, the two Western Movie course teachers were invited to evaluate the lists. After consultation, they settled on 17 and 19 background elements and 19 and 16 narrative elements for Video 1 and Video 2, respectively. Establishing these lists allowed for the extraction of video-based elements from the students' brainstormed ideas. To start, text was extracted from the mind maps submitted in picture format using Editpad, a free and open-source image-to-text converter. Afterward, using the Beyond Compare software, the text files were aligned, and identical words and expressions were identified. The aligned texts were also manually checked to ensure that synonymous expressions were identified and extracted as necessary. To investigate the differences between the two groups, we calculated the M score and SD. This was followed by an independent sample t-test, and the statistical differences were found based on the p-value. The practical differences, based on Cohen's d-values, were also obtained.

Additionally, the participants' notes were analyzed to determine how often the EG rewatched the videos during the BS process. Using Sketch Engine, a small corpus of participants' notes was created and analyzed. The Keyword function of the corpus analysis tool was used to extract words associated with "watch," "rewatch," and "video" to determine who rewatched which video and how often. The data from the two groups were compared again using an independent t-test after calculating the M score and SD of each group. Additionally, Pearson's correlation coefficient was computed to determine the relationship between the number of video (re)watches and the ideas generated. In each case, the p-value and Cohen's d-value were obtained to determine the statistical and practical differences (t-tests), and the p-value was computed to determine the correlation coefficients.

RQ3 was answered by analyzing the questionnaire responses. In particular, the means and standard deviations of the questionnaire responses were calculated to determine the participants' perceptions of the videos' usefulness, satisfaction with their quality and sound, and perceived benefits. The M scores were calculated, and a one-way analysis of variance (ANOVA) was performed to determine the statistical differences between the various perceptions. Furthermore, Pearson's correlation coefficients were computed to determine whether the variables had any bearing on the brainstormed ideas generated by the EG.

4 Findings

4.1 Research question 1: Did the AI-generated videos foster idea generation during brainstorming?

4.1.1 Ideas generated

Our analysis revealed that across both BS sessions, the EG generated a total of 1,118 ideas, surpassing the CG, which produced 1,031 ideas. However, the data indicated that both groups generated more ideas during BS1 (EG: $n = 629$; CG: $n = 532$) than during BS2 (EG: $n = 489$; CG: $n = 499$). Notably, the CG generated more ideas during BS2 than the EG. In terms of percentages, the CG generated 52% of its ideas in BS1 and 48% in BS2, while the EG generated 56% of its ideas in BS1 and 44% in BS2.

The analysis of the means and standard deviations of both groups is presented in Table 1.

Despite the nonexistence of statistical differences, we noted that both groups maintained slightly dissimilar idea generation patterns. While the two groups generated significantly more ideas at BS Levels 2 and 3 (CG: 42.68% and 30.26%, respectively; EG: 36.58% and 26.21%, respectively), the data indicated that the EG generated more ideas at BS Levels 4 and 5, as presented in Fig. 5.

Table 1 Mean (M) scores and standard deviations (SD) of the ideas generated by the control and experimental groups

Level	Control group	Experimental group	p-value
Total ideas	M = 44.83; SD = 20.15	M = 48.13; SD = 21.94	p = 0.62 (no statistical difference)
Level 1	M = 8.65; SD = 2.96	M = 9.57; SD = 6.59	p = 0.97 (no statistical difference)
Level 2	M = 19.73; SD = 10.31	M = 17.48; SD = 7.10	p = 0.48 (no statistical difference)
Level 3	M = 13.56; SD = 11.50	M = 12.74; SD = 9.20	p = 0.76 (no statistical difference)
Level 4	M = 3.39; SD = 3.39	M = 7.35; SD = 12.47	p = 0.14 (no statistical difference)
Level 5	M = 0.09; SD = 0.42	M = 1.00; SD = 2.22	p = 0.07 (no statistical difference)

The BS pattern of the EG indicated that the participants generated more sub-ideas while focusing on details that built on ideas from previous levels—a think-through BS strategy. Notably, even though the CG generated more ideas during BS2, the EG's ideas were more spread out among the levels. In other words, while the videos may not have contributed to the generation of more ideas by the EG, statistically speaking, they may have allowed these participants to generate more sub-ideas, thereby broadening their thinking patterns.

Our analysis of the ideas generated by each participant revealed significant variability in the data. On average, the participants in the EG produced 48.13 ideas, while those in the CG generated an average of 44.83 ideas. However, this average masked considerable disparities among the individual contributions. Some participants generated far more ideas than others, with the lowest number being 15 for both groups. By contrast, the highest outputs were 92 ideas for the EG and 81 for the CG, as illustrated in Fig. 6.

Many participants in the EG yielded less than the average number of ideas, given that 43.4% generated more than 48 ideas in the entire class. Meanwhile, 52.2% of the participants in the CG generated more than the group's average of 44.83 ideas.

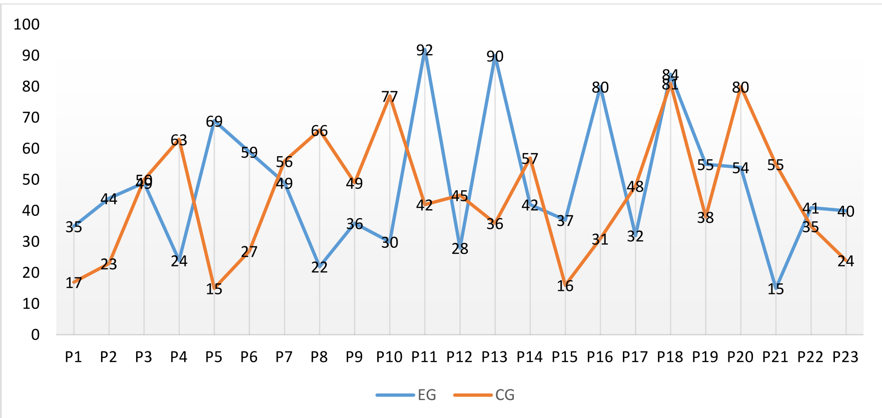


Fig. 6 Range of ideas generated by the control group (CG) and experimental group (EG)

While individual BS skills may account for variations in idea generation, we aimed to assess the extent to which the videos influenced the participants in the experimental group (EG) by correlating their frequency of video rewatching with the number of ideas generated. The participants' notes documented how often they revisited the videos during the BS process. Our analysis revealed that those who generated a higher number of ideas tended to have watched the videos more frequently than their peers who produced fewer ideas. For instance, participants 11 (P11), P13, P16, and P18, each generating more than 80 ideas, reported watching the videos five or six times. We employed SPSS to calculate the Pearson correlation coefficient between the number of ideas generated and the frequency of video rewatching. The results indicated a strong positive and statistically significant correlation [$r(23)=0.89$; $p=0.001$], suggesting that the frequency of video rewatching significantly influenced the number of ideas generated.

4.1.2 Time (not) spent on the brainstorming task

Our analysis unveiled no statistical differences between the time (in seconds) that the EG and CG spent on the BS task, as demonstrated in Table 2.

However, the CG spent less time on BS1 (34%) and more time on BS2 (66%). The group had more inactive time in BS1 (54%) than in BS2 (46%), and in terms of actual BS time, the CG participants spent an average amount of time on BS2 (51%) and BS1 (49%).

Overall, the EG spent more time on BS1 (55%) than on BS2 (45%). In terms of inactivity time, the EG participants spent more time on BS1 (62%) than on BS2 (38%) and slightly more time on BS2 (52%) than on BS1 (48%). The data indicated that both groups only spent approximately half of their time on actual BS. We attempted to understand the variation in the data by analyzing the participants' notes and the individual time spent on the BS activities. Regarding the individual time spent on BS, we found no significant differences between the top performers in the EG (except P13) and CG, as indicated in Fig. 7.

We also investigated whether the time the EG spent on the task affected the number of times the participants watched the videos and found no correlation ($p=0.79$). Furthermore, we found no correlations between the number of ideas generated and the total time spent on the BS task for the CG ($p=0.87$) and the EG ($p=0.72$).

However, we found significant differences in the BS patterns adopted by the participants, showing the existence of differences between the two groups. For instance, we compared the time spent on BS with the number of ideas generated by the CG

Table 2 Time spent on the brainstorming task by the control group and experimental group

Time	Control group	Experimental group	p-value
Total time	$M=913.65$; $SD=358.40$	$M=1084.48$; $SD=542.05$	0.22 (not significant)
Inactivity time	$M=412.48$; $SD=258.00$	$M=513$; $SD=554.91$	0.35 (not significant)
Actual time spent brainstorming	$M=501.17$; $SD=264.12$	$M=571.47$; $SD=185.01$	0.33 (not significant)

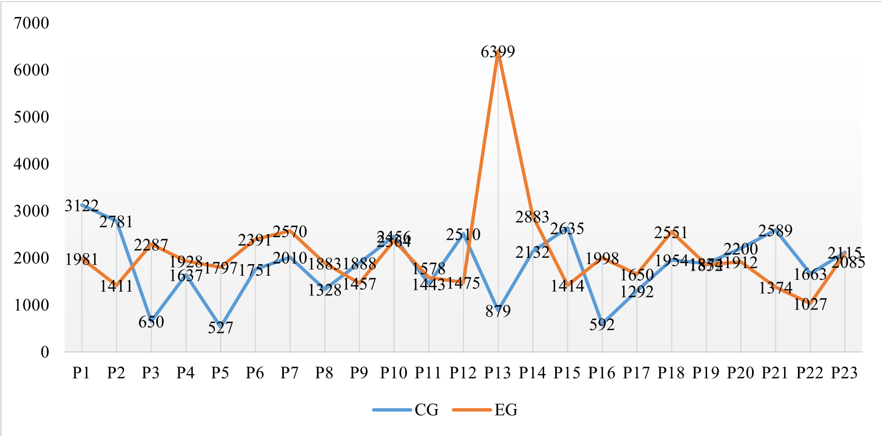


Fig. 7 Differences in individual performances between the control group (CG) and experimental group (EG)

and EG. With regard to the CG, we found that not everyone who spent over 2,400 s on the task generated the most ideas, as indicated in Table 3.

Notably, P1 and P2 generated few ideas, though they spent a significant amount of time on the BS task. We checked their notes and found that they were predominantly "thinking about ideas and consulting online resources." However, when we rewatched their videos, we found that they spent a significant amount of time typing their ideas in English, their second language. We also found that, most often, they forgot to change the keyboard options to English, meaning each time they typed a word, their keyboard brought up Pinyin and Chinese characters instead of English characters, as highlighted in Fig. 8.

We also examined the EG participants who spent over 2,400 s on both tasks. Our findings revealed that, in contrast to the CG, the majority of these participants generated a significantly higher number of ideas, as illustrated in Table 4.

Upon rewatching P7's screen recording, we uncovered a few practical elements that used up the participant's BS time. Contrary to the instruction and previous

Table 3 Control group participants with the longest brainstorming (BS) time and the ideas they generated

	Time total (in seconds)	BS1 time (in seconds)	Ideas generated in BS1	BS2 time (in seconds)	Ideas generated in BS2
P1	3,122	934	8	2,188	9
P2	2,781	1,704	11	2,781	12
P10	2,456	1,114	38	2,456	39
P15	2,635	1,541	16	2,635	20
P21	2,589	1,232	20	2,589	55

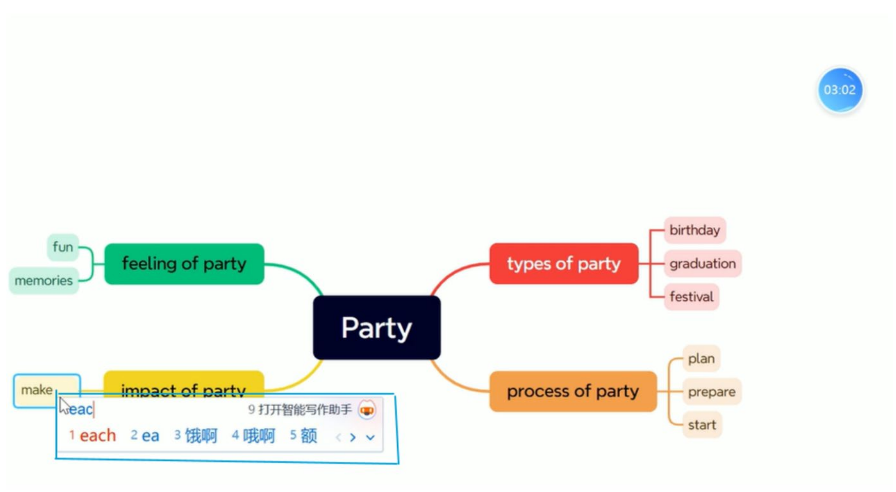


Fig. 8 Screenshot showing how the keyboard typing options influenced the participants' speed

Table 4 Experimental group participants with the longest brainstorming (BS) time and the ideas they generated

Participant	Total time (in seconds)	BS1 time (in seconds)	Ideas generated in BS1	BS2 time (in seconds)	Ideas generated in BS2
P7	2,570	1,610	30	960	19
P13	6,399	5,448	53	951	37
P18	2,551	938	46	1,613	28

practice, P7 used many words, mostly phrases and clauses, to represent the ideas generated. For instance, they predominantly used the article "the" and clauses instead of a single word or phrase, as indicated in Fig. 9.

The participants were instructed to avoid articles and to use as few words as possible during the BS to save time and generate more ideas.

4.2 Research question 2: Which elements of the artificial intelligence–generated videos were most explored by the learners?

4.2.1 Presence of artificial intelligence–generated video background elements in the ideas generated

Video 1 A total of 17 GenAI T2V background elements were extracted from Video 1 and analyzed to address this research question. We examined the mind maps of the participants from the CG and EG to assess how the AI-based videos' backgrounds

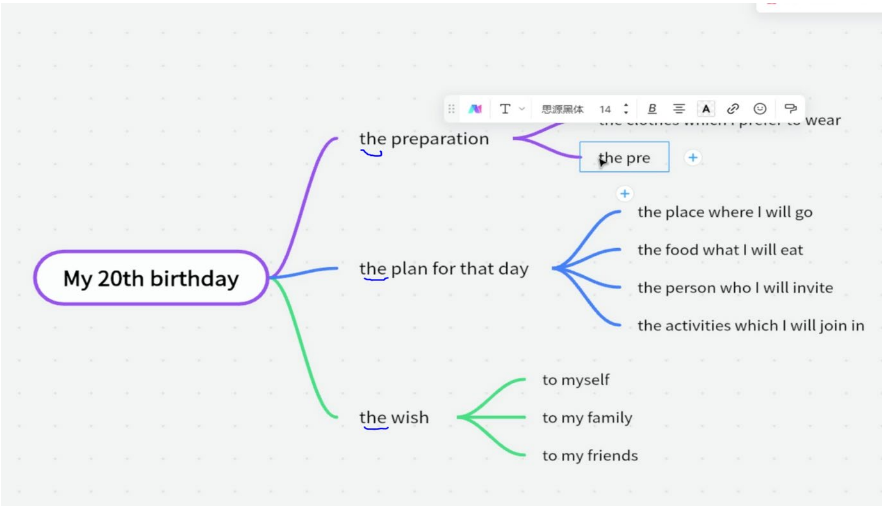


Fig. 9 Screenshot showing the presentation of Participant 7's generated ideas

facilitated idea generation among the EG participants. Our analysis of the first essay topic, "My 20th birthday," revealed that the ideas produced by the EG more closely aligned with the background features of the AI-generated video compared with those generated by the CG, which did not view the videos prior to BS. Specifically, the EG generated 66 ideas that corresponded with the video's background elements, while the CG produced only 42 aligned ideas. This finding indicates that the EG generated 36.36% more ideas than the CG, as illustrated in Fig. 10.

The most significant differences concerned Elements 12 (richly decorated cake) and 1 (multiracial attendees), which were mentioned by up to 7 and 6 more

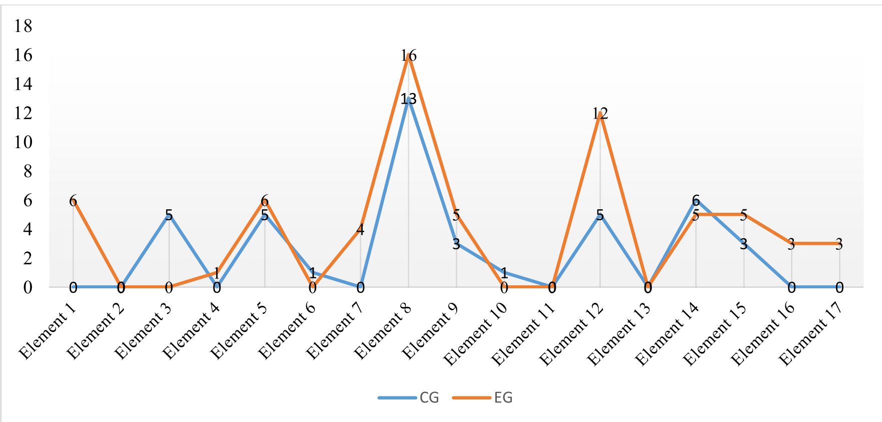


Fig. 10 Comparison of the alignment of the control group's (CG) and experimental group's (EG) generated ideas with artificial intelligence–generated Video 1's background elements

participants, respectively, in the EG. Other differences involved Elements 8 (friends and family), 16 (evening party), and 17 (beach scene). A significant difference was revealed regarding Element 3 (burning candles), which was mentioned by more CG participants (n = 5) than EG participants (n = 0).

Furthermore, we computed the data in SPSS and performed a t-test to determine the statistical differences between the two groups' data. A significant statistical difference was present between the EG's (M = 3.67; SD = 4.47) and the CG's (M = 2.33; SD = 3.44) data, with a p-value of 0.03. A large effect size was also discovered, judging by a Cohen's d value of 2.81 (significantly large).

Video 2 Regarding Video 2 ("A beach party I organized"), 19 background elements were identified and compared with the BS ideas generated by the EG and CG. As with Video 1, we found that the participants in the EG generated more ideas (n = 77) than the participants in the CG (n = 66), suggesting that the T2V input did not translate into more ideas for the EG. As indicated in Fig. 11, some differences in the individual ideas were noteworthy in the data of the two groups.

The participants in the EG were more influenced by Elements 3 (people dancing), 10 (stars and sun), 11 (boys and girls splashing water), and 15 (adults drinking), while those in the CG did not yield significantly more ideas (3 +) in any category. However, despite these apparent differences, a sample t-test revealed no significant statistical differences between the two groups, meaning that the EG did not significantly benefit from the GenAI-based T2V support.

Finally, the EG's ideas generated during BS1 and BS2 were introduced into SPSS to investigate the existence of any correlation with the total ideas generated during the BS. Our analysis found no correlation, meaning that the videos' background elements did not significantly influence the total ideas generated during both BS sessions.

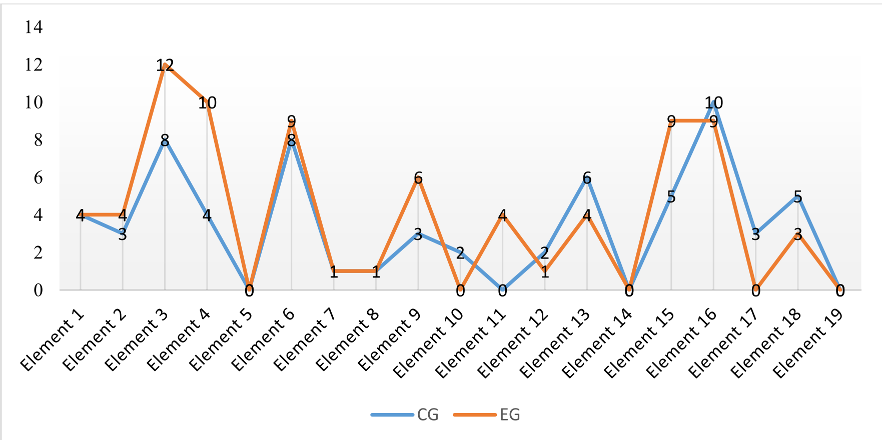


Fig. 11 Comparison of the alignment of the control group's (CG) and experimental group's (EG) generated ideas with artificial intelligence–generated Video 2's background elements

4.2.2 Presence of artificial intelligence–generated video narrative elements

In addition to the background elements of the AI-based videos, we examined the narration or narrative elements underpinning the videos. The exclusively AI-generated narration contained multiple elements, including adjectives, phrases, and clauses, to trigger ideas during BS.

4.2.3 Video 1

In total, 19 narrative elements were identified in Video 1, including 11 (57.89%) coinciding with the CG's brainstormed ideas and 8 (42.11%) coinciding with the EG's brainstormed ideas. This means the CG's brainstormed ideas aligned more with the GenAI-based video than those of the EG, which watched the video and was presumed to outperform the CG. Overall, 24 narrative elements were utilized by the CG as opposed to 16 employed by the EG. Therefore, the CG utilized five more narrative elements (31.25%) than the EG. The individual differences are outlined in Fig. 12.

Individually, we found significant discrepancies in both groups in terms of narrative elements employed, as indicated in Fig. 9. The data showed that the ideas generated by participants in the EG aligned more with Elements 7 (heartfelt laughter), 11 (shared stories), and 17 (toast/cheers to a new chapter), while those in the CG aligned more with Elements 2 (sun in the horizon) and 6 (buzzing with excitement). However, despite the differences in the alignment of the narrative elements, a sample t-test found no statistical differences between the two groups.

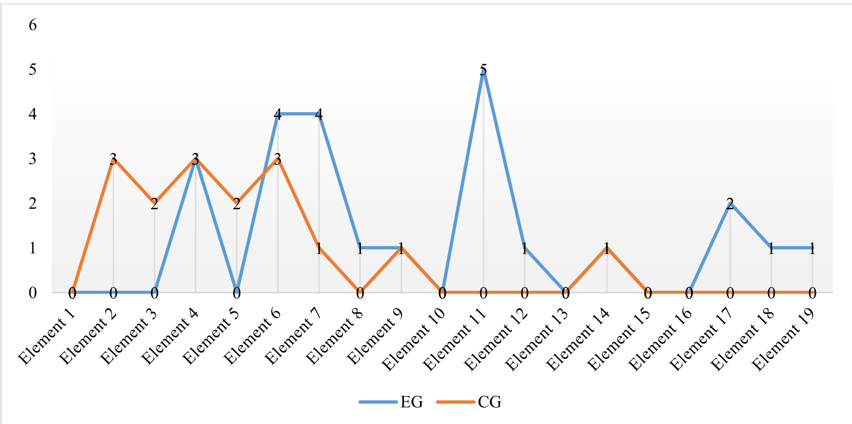


Fig. 12 Differences in the control group's (CG) and experimental group's (EG) brainstormed ideas compared with Video 1's narrative elements

4.2.4 Video 2

As with Video 1, we identified and extracted narrative elements of the AI-generated video that we presumed could impact ideation by the EG participants. In total, 16 elements (individual adjectives, noun phrases, clauses, and other expressions) were identified. Our analysis indicated that the ideas generated by the EG conformed more to the AI-generated video's narrative elements compared with those of the CG. Specifically, the EG's brainstormed ideas aligned 18 times with the AI-generated video (M = 4.05; SD = 3.99). The CG's brainstormed ideas aligned only 9 times (M = 3.42; SD = 2.99), meaning that 50% fewer of their ideas aligned with the AI-generated video's narrative elements. As indicated in Fig. 13, significant differences were present in the elements chosen by the participants of the two groups.

The data analysis indicated that the ideas generated by the EG participants aligned more with Elements 1 (sunshine kisses your skin) and 6 (unforgettable day) compared with those of the CG participants. Moreover, the ideas generated by the participants in both groups did not align with Elements 2 (serene sea waves), 3 (idyllic spot), 4 (golden sands), 8 (party guests), 9 (completely transformed), 10 (tables adorned with tropical flowers), 11 (DJ booth), or 15 (zenith). We also found that more elements (5 out of 16; 31.25%) did not align with any ideas generated based on Video 2 as opposed to those generated based on Video 1, where 5 out of 19 elements (26.32%) did not align with any ideas generated by the participants in the two groups. As with Video 1, we also found no significant statistical differences between the data of the EG and CG.

Furthermore, we analyzed the individual data of the EG participants to determine whether the videos influenced some participants more than others, whether statistical differences existed in the ideas generated during BS1 and BS2, and whether correlations were present between the background and narrative elements and the total ideas generated. We found that more ideas generated during BS2 aligned with

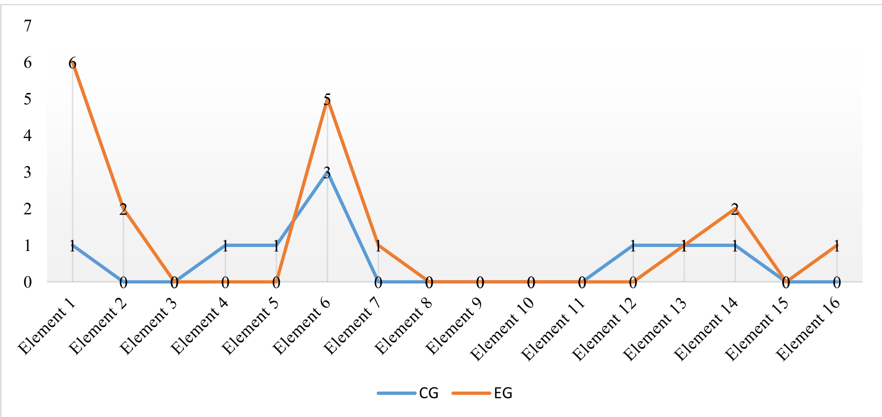


Fig. 13 Differences in how the elements of Video 2 aligned with the brainstormed ideas of the control group (CG) and experimental group (EG) participants

the narrative elements of Video 2 (20) as opposed to those of Video 1 (16) and that more participants' BS2 ideas (56.52%) aligned with Video 2's narrative elements than their BS1 ideas for Video 1 (34.78). No participant seemed to have depended on the AI-generated videos to any significant extent, as indicated in Fig. 14.

The data indicated an inconsistent pattern wherein some participants' ideas aligned with one video but not the other. For instance, the ideas generated by P2 during BS1 were not aligned in any way with Video 1, but during BS2, two ideas they generated aligned with the narrative elements of Video 2. This pattern was also common among P5, P8, P9, P10, P13, P15, P16, P17, P19, P21, P22, and P23 (i.e., 13 [56.52%] of the participants).

4.3 Research question 3: What were the learners' perceptions regarding the artificial intelligence–generated videos for brainstorming?

To answer RQ3, we analyzed the EG participants' perceptions of the videos' usefulness and their satisfaction with the image quality and narration. We also investigated whether such perceptions correlated with the ideas generated.

4.4 Better impression of the artificial intelligence–generated video's usefulness and quality

Our analysis of the questionnaire responses' M score and SD determined that the participants believed that the AI-generated videos were useful and of high quality. Furthermore, they perceived the videos' narrative elements as satisfactory. Comparatively, the narrative elements had a lower M score, as presented in Table 5.

A one-way ANOVA was conducted to examine the differences in the students' perceived usefulness, satisfaction, and benefit of AI-generated videos for idea creation. The analysis revealed significant differences among the groups [$F(2, 66) = 18.593$; $p < 0.001$,] indicating that the students' perceptions varied significantly

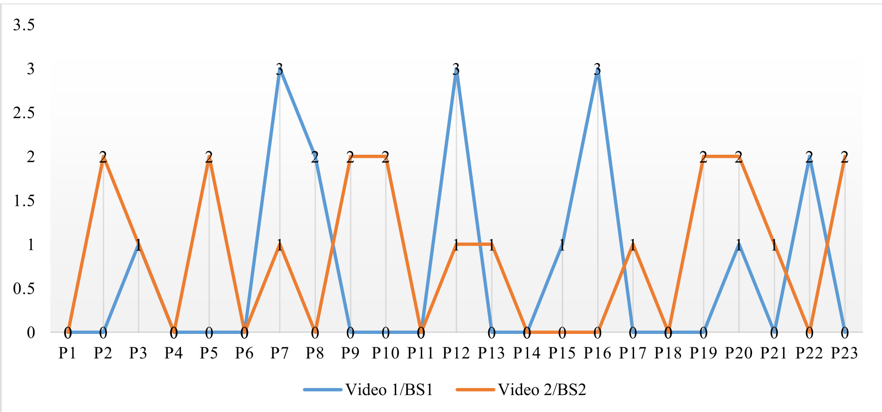


Fig. 14 Extent to which the experimental group's ideas generated in brainstorming 1 (BS1) and brainstorming 2 (BS2) aligned with the artificial intelligence–generated Video 1 and Video 2

Table 5 Mean score and standard deviation of the experimental group participants' perceptions of the artificial intelligence (AI)-generated videos

Item	Mean	Standard Deviation
The overall usefulness of te AI-generated videos	4.61	0.50
Satisfaction with AI-generated video quality	4.78	0.42
Satisfaction with AI-generated video narration	3.78	0.80

based on the group. The findings suggested that the students perceived the AI-generated videos as highly useful and satisfying tools for idea creation, though the perceived benefits were somewhat lower. The significant ANOVA results indicated that the differences in perceptions among the groups were meaningful, and the large effect sizes implied that the AI-generated videos had a substantial impact on students' experiences related to idea generation.

The effect sizes from the analysis provided important insights into the impact of group membership on the perceived usefulness, satisfaction, and benefits of the AI-generated videos. Eta-squared was 0.360, indicating a large effect size and suggesting that approximately 36% of the variance in the dependent variables could be attributed to group membership. Similarly, Epsilon-squared was 0.341, reinforcing the large effect size observed in Eta-squared. The fixed-effect Omega-squared value of 0.338 further supported this by pointing to a reliable estimate of the proportion of the variance attributable to the treatment effect. By contrast, the random-effect Omega-squared estimate of 0.203 reflected a moderate effect size, which could suggest variability within the population. Together, these effect size measures highlighted the substantial impact of the AI-generated videos on the students' perceptions.

Finally, our analysis revealed no correlations between the participants' perceptions of the videos (i.e., usefulness, quality, and narration) and other variables, including the total BS ideas generated and video background and narrative elements. The findings proved that the participants' perceptions regarding the usefulness and quality of the videos did not translate into concrete ideation during the BS process.

5 Discussion

The study's findings were very revealing regarding the current and future applications of AI-generated videos in FL learning, particularly during BS. First, concerning whether the AI-generated videos could foster idea creation, we found no statistical differences between the EG and the CG. However, our analysis indicated that the EG participants generated more main ideas and sub-ideas, especially at Levels 4 and 5, than the CG participants. Second, we found correlations between the number of times the AI-generated videos were watched and the number of ideas generated by the EG. Third, no statistical differences were evident between the EG and the CG regarding the time spent on the BS task. Nevertheless, we discovered that individual practices related to keyboard setting, typing speed, and compliance with instructions negatively impacted specific participants' BS time. Fourth, the analysis of the participants' mind maps showed that the ideas generated by the EG aligned more with the AI-generated T2V background elements than those generated by the CG. Fifth, the BS ideas generated by both groups aligned with the narrative elements of the videos, and no statistical differences were found between the two outcomes. Sixth, the data analysis indicated that the participants had a higher perception of the usefulness and quality of the videos than their narrative elements, even though such perceptions had no correlations with the BS ideas generated by the EG. We also established that the participants' BS ideas differed according to the video they watched.

These findings contribute to the literature on AI-based T2V models in education by providing empirical evidence regarding their role in enhancing BS activities among college students. This study supports previous research by Liu et al. (2024), AlAfnan et al. (2023), Ariyaratne et al. (2023), and Giray (2023) speculating on the importance of AI-generated videos in education, particularly in writing. It aligns with Yu-Han and Chun-Ching's (2023) conclusion that learners have favorable perceptions of such videos and emphasizes the need to consider individual attitudes in learning contexts. However, the results do not sufficiently support the claim that AI-generated videos can extend human thinking in BS, suggesting that this conclusion may be premature. The statistical non-differences between the EG and the CG highlight the novel nature of AI-generated videos and their pedagogical applicability. Overall, the study underscores the importance of participants' attitudes in reassessing the impact of GenAI T2V models in FL learning, with implications for practical, pedagogical, theoretical, and policy aspects of language teaching.

One of this study's practical implications is that GenAI T2V models can significantly enhance BS and creative processes in academic settings. The notable increase in idea generation among the EG indicates that educators should integrate these tools into their curricula to assist students in generating ideas for writing and other projects. Furthermore, the positive correlation between video rewatching and idea generation connotes that encouraging students to engage with multimedia resources multiple times can lead to better outcomes (Bašić et al., 2023; Chen & Wu, 2024). Strategies that promote rewatching, such as structured reflection sessions and guided discussions, may be effective. Additionally, the findings highlight the need for high-quality, engaging multimedia resources tailored to specific academic tasks, conveying that educational institutions should invest in creating or curating relevant videos. Professional development focused on integrating GenAI tools may enhance teaching practices (Ding et al., 2024), and monitoring student engagement with these videos can provide valuable insights into learning processes. Moreover, differences in idea generation patterns between the EG and the CG underscore the necessity to refine instructional strategies to ensure that students are supported in effectively utilizing multimedia resources.

The pedagogical implications emphasize the significant role of GenAI T2V models in enhancing BS and creativity. Educators should incorporate these tools into their curricula to support students' idea generation, especially during writing instruction. The positive correlation between video rewatching and idea generation also suggests that instructors may encourage repeated engagement with multimedia resources, promoting deeper content exploration. Structuring assignments that require rewatching videos followed by reflective discussions may be beneficial. Moreover, the study underscores the importance of developing engaging multimedia resources aligned with specific learning objectives. Educators are encouraged to curate or create videos that stimulate creativity effectively (Leiker et al., 2023). Professional development for educators is crucial to help them learn how to incorporate GenAI tools into their teaching strategies (Karanjakwut & Charunsri, 2025). Understanding variations in idea generation patterns can inform instructional approaches, enabling targeted support for all students (Mahapatra, 2024). Overall, these implications highlight the potential of GenAI technologies to enhance pedagogical practices and foster creativity.

The theoretical implications are also noteworthy. This study enhances our understanding of how multimedia elements, particularly AI-generated videos, influence cognitive processes during BS, thus contributing to the literature on creativity and cognitive engagement in writing tasks (Lee, 2020). The study reinforces Alobaid's (2020) study, which found that multimedia resources are effective and are strongly recommended "for language learners and teachers where optimization of writing fluency is the target of learning" (p. 1) The findings also suggest that multimedia resources can significantly impact student engagement and idea generation (Romero-Hall et al., 2020), prompting further exploration of how different media types interact with cognitive processes. Additionally, by demonstrating the effectiveness of GenAI tools in BS, this research supports the need for updated theoretical models that incorporate technology's role in academic writing, indicating a shift in how writing processes are conceptualized. The variability in individual responses to multimedia stimuli highlights the need for theories related to personalized learning and adaptive instructional approaches in writing education in future research—an idea echoed by Karanjakwut and Charunsri (2025).

The study's findings suggest important policy implications for enhancing educational practices. Academic institutions should develop policies that encourage the integration of GenAI tools into the curriculum to improve BS and writing processes. Policymakers should allocate funding for developing and implementing AI technologies in educational settings, ensuring that schools and universities have the necessary resources. Frameworks such as the "AI Ecological Education Policy Framework to address the multifaceted implications of AI integration in university teaching and learning" (Chan, 2023, p. 1) should be implemented to bring together various stakeholders involved in AI implementation in higher education learning. Additionally, policies supporting ongoing professional development for educators are essential to equip them with the skills needed to integrate GenAI technologies effectively. Establishing guidelines for the use of AI tools in education will help educators understand best practices and address potential challenges (Leiker et al., 2023). Finally, promoting research and evaluation initiatives to assess the impact of AI tools on learning outcomes will inform future policy decisions and refine technology integration in education.

6 Conclusion

This study investigated the use of an AI-generated T2V model in idea creation by college students taking a writing course. The data obtained from the questionnaire, screen recordings, mind maps, and participants' notes were analyzed to determine whether the EG spent less time than the CG on generating more ideas by exploring the narrative and background elements of AI-generated videos. According to the findings, no statistical differences were found between the two groups concerning the time spent on the BS task and the ideas generated. However, the findings revealed that the number of ideas generated correlated with how often the participants watched the videos, with the participants spending an unequal amount of time on each video. Furthermore, we found that the BS time hinged on the participants' typing speed, keyboard setting, and compliance with the instructions. The questionnaire analysis determined that the EG participants better perceived the usefulness and quality of the AI-generated videos rather than their narrative elements.

Based on the findings, concluding that GenAI T2V models trigger more idea generation during BS was deemed premature. Contributing to expanding the discourse on AI's integration into language learning, especially in college-level writing, the study has several practical implications. With the help of instructors, students need to learn how to fully maximize AI-generated video elements (background and narrative) during BS.

The novelty of this study lies in its empirical evidence demonstrating the impact of GenAI T2V models on idea generation in BS activities, particularly within FL learning contexts. While previous research has speculated on AI's educational role, this study provides concrete data from college students, revealing a significant correlation between video rewatching and the number of ideas generated and suggesting that repeated engagement with multimedia resources enhances cognitive processing and creativity. Additionally, the study examines individual factors, such as keyboard settings and typing speed, which influence BS outcomes, contributing to a nuanced understanding of student interactions with technology. By highlighting the effectiveness of GenAI tools in BS, the research supports the need for updated theoretical frameworks that incorporate technology's role in academic writing and creativity. Furthermore, the study outlines comprehensive implications—practical, pedagogical, theoretical, and policy-related—underscoring the transformative potential of AI technologies in educational practices.

7 Limitations

The study participants included two freshmen classes (CG and EG), each comprising 23 learners. Though this research was representative, a study with more participants may offer new insights and perspectives regarding BS in academic writing. Moreover, the study focused on individual participants working independently. Therefore, the findings do not apply to participants working collaboratively in groups and exchanging ideas during such a task. Furthermore, only two AI-generated videos were examined in this study; future researchers may explore more videos with varied contents and narrative elements. This study examined ideas generated during BS without investigating whether such ideas made their way into the essays written by the participants. Future investigations of this relationship may complement the findings of this study or offer novel evidence on the impact of AI-generated videos in idea creation and essay writing. Additionally, the diversity and usefulness of the ideas generated were measured by assessing how well they aligned with the AI-generated videos' narrative and background elements; future studies may employ other assessment methods to triangulate the findings of this study.

Appendix 1. Writing course schedule for the semester

Week 1 and 2- Introduction to Descriptive Essays
- Description vs. Narration
- Descriptive Essay Structure

Week 3 and 4- The Descriptive Essay Writing Process
- Brainstorming

Week 5 and 6 - Brainstorming (individual)
- Brainstorming (group)

Week 7 and 8 - Structuring ideas into paragraphs
- Adding ideas to complete the essay structure
- Ensuring there is sensory information
- Ensuring the story is interesting

Week 9 and 10- Writing the introduction
- Homework (individual and group work)
- Feedback

Week 11 and 12- Writing body paragraphs
- Homework (individual and group work)
- Feedback

Week 13 and 14 - Writing the conclusion
- Homework (individual and group work)
- Feedback

Week 15 and 16 - Further practice with paragraphs
- Writing a full essay
- Feedback

Week 17 and 18 - Revision
- End-of-semester project writing

Acknowledgements N/A

Funding Funding for this project was obtained from Fujian Provincial Social Science Fund Major Project, No. FJ2022XZZ007.

Data availability Data generated for the purpose of this research is available with the corresponding author upon reasonable request.

Declarations

Competing interests No competing interest to declare.

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